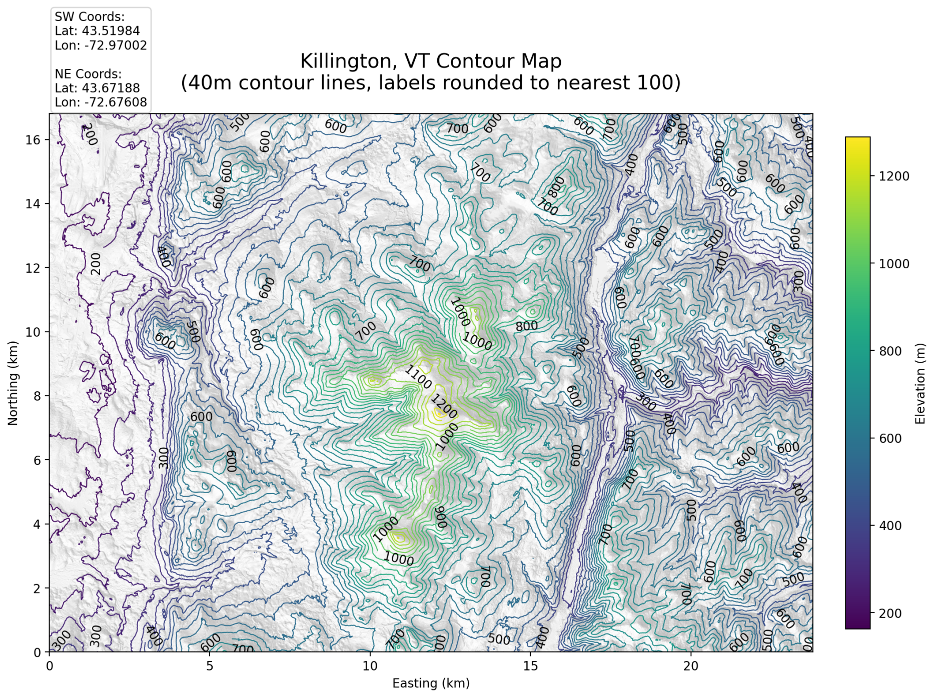
Spatial Stat Assignment 8

GEORGE BASTA, LUKE ANDRADE, PRANAV TIKKAWAR

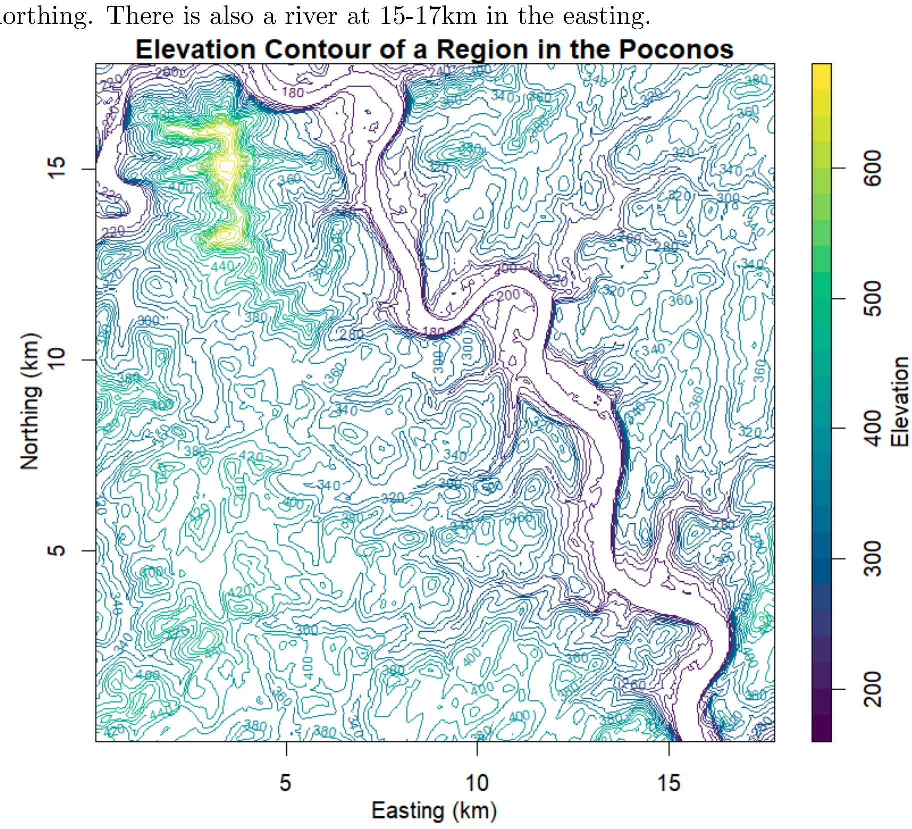
April 2026

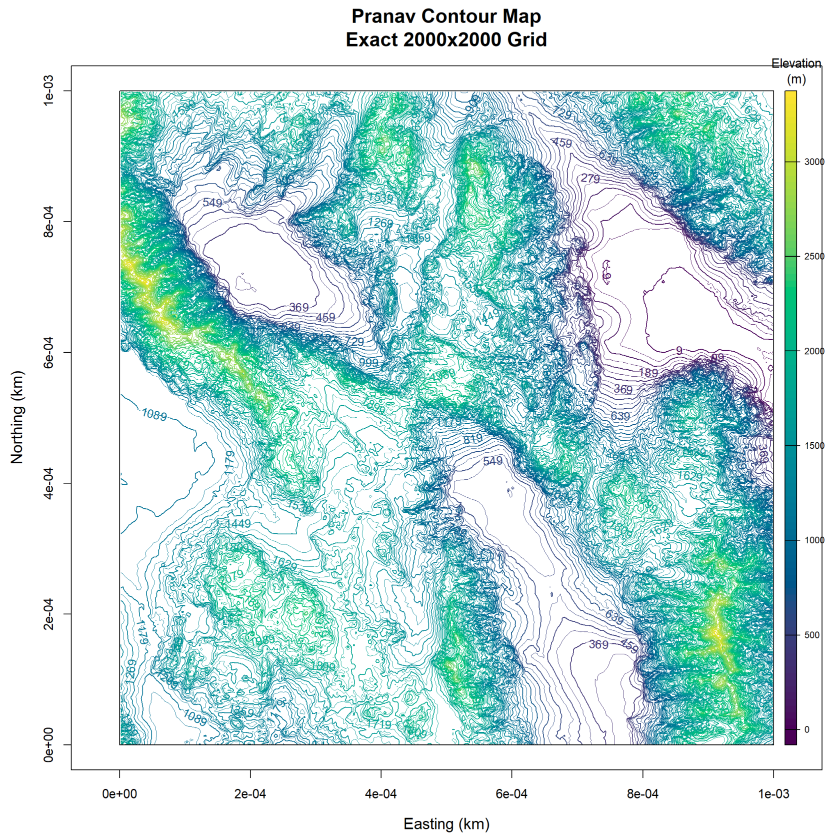
Contour Plots of Regions

The plots in the following sections will be in the same order as these contour plots.



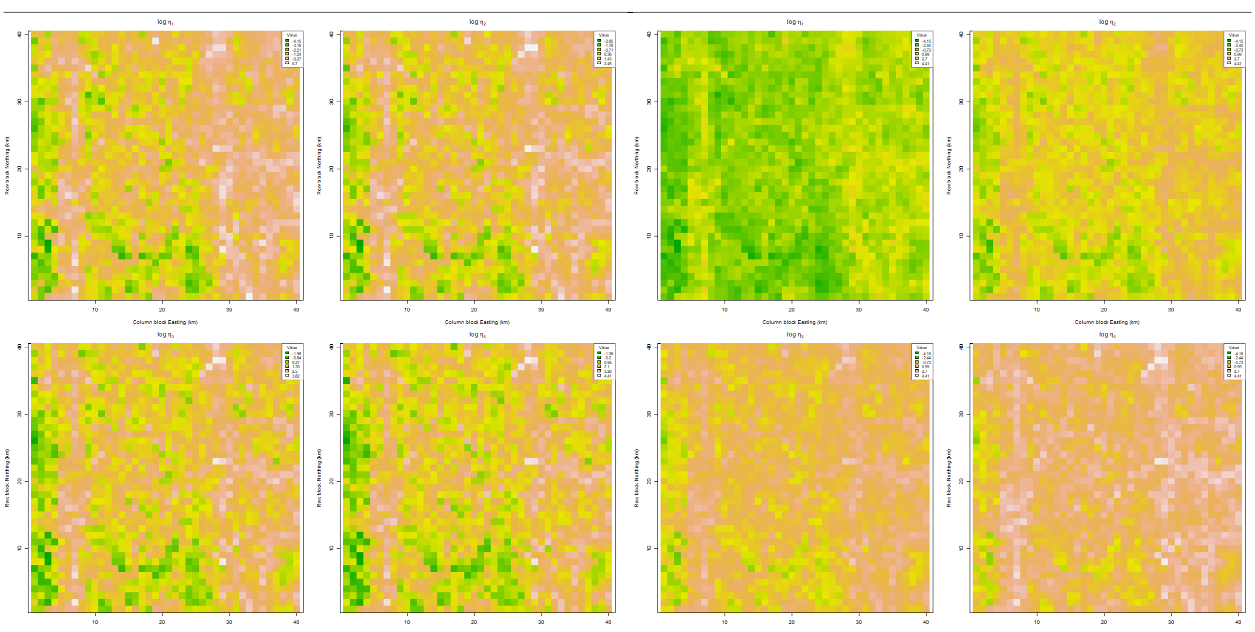
Killington, VT notable features are that there is a mountain around 10-15km in the easting and 4-10km in the northing. There is also a river at 15-17km in the easting.





This is a section of the Death Valley region that varies drastically in elevation in many different areas.

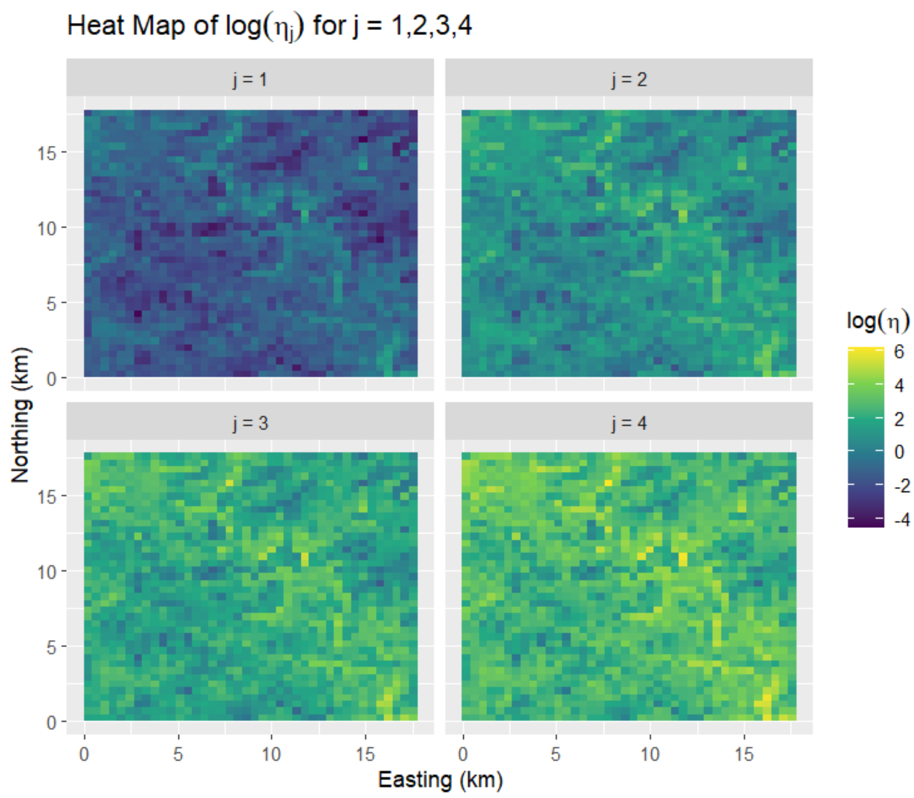
Heat Map of η_j and $\log \eta_j$ For $j = 1, 2, 3, 4$



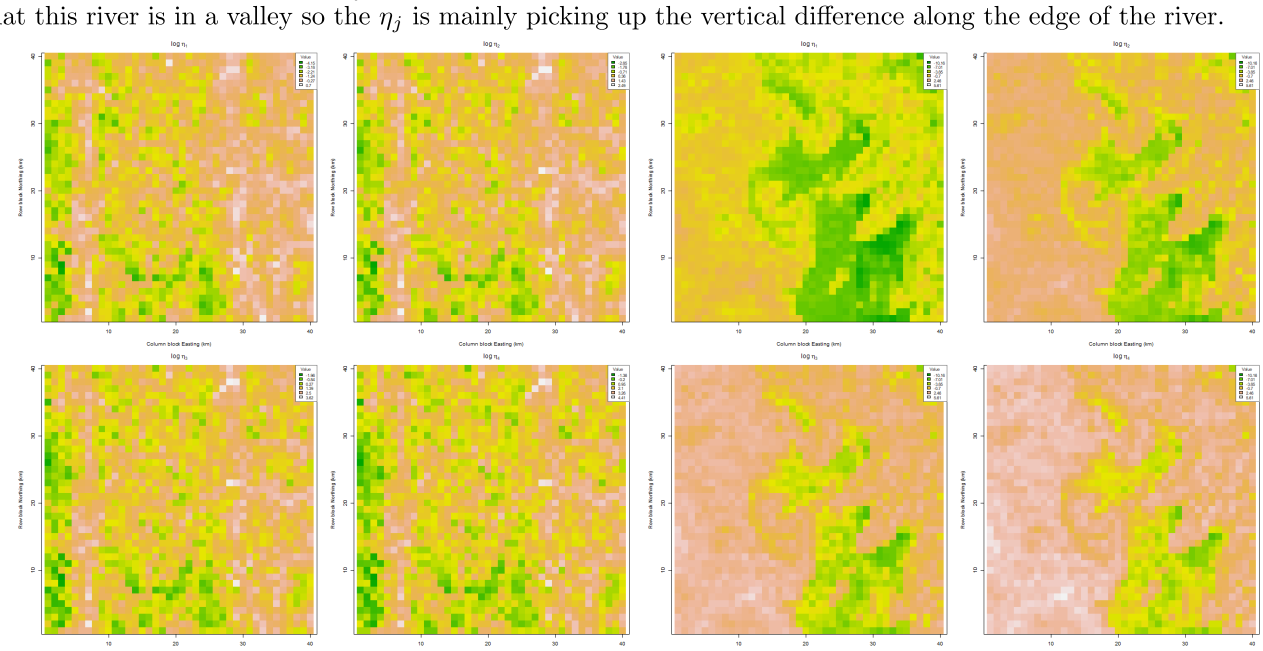
In the η_j maps here, the left four use a local range for each plot while the right four use a global range. The legend is small but this is for 50x50 pixel blocks and the darkest green corresponds to -4.15 η_j values all the way up to 4.41 for white blocks using global range, while the local ranges vary from $\log \eta_1$ having the lower values in that global range and $\log \eta_4$ having the higher values. η_1 is the top left, η_2 top right, η_3 bottom left and η_4 bottom right.

I'm not really sure how to interpret the local plots, they seem to be giving the same plot

In the global plots for η_1 to η_4, it does appear that amplitude of the spatial patterns are increasing and getting more obvious as the lag increases, but the patterns are nearly identical across $j = 1, 2, 3, 4$.
Around the river there are the highest $\log \eta_j$ across all j and this shows the river is not very smooth and varies a lot, and there is probably strong anisotropy around this river since we are in Vermont. Also, near the left-center of the $\log \eta_j$ has higher values implying the approach to the mountain is rougher (the contours around this region are also very tight which is further indication). A more numerical analysis will come next.

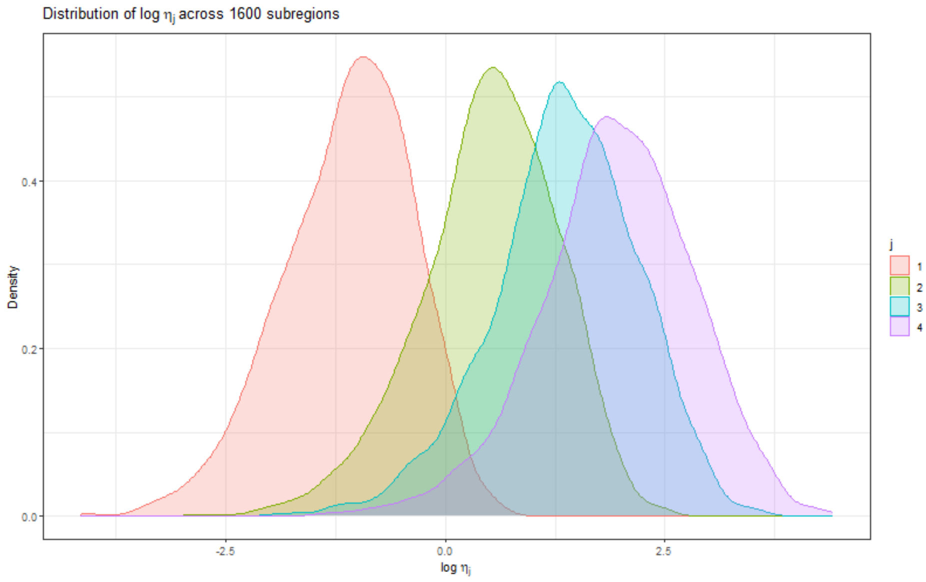


Based on the heat map, the η_j is a lot higher on the river compared to the rest of the area. This indicates that this river is in a valley so the η_j is mainly picking up the vertical difference along the edge of the river.



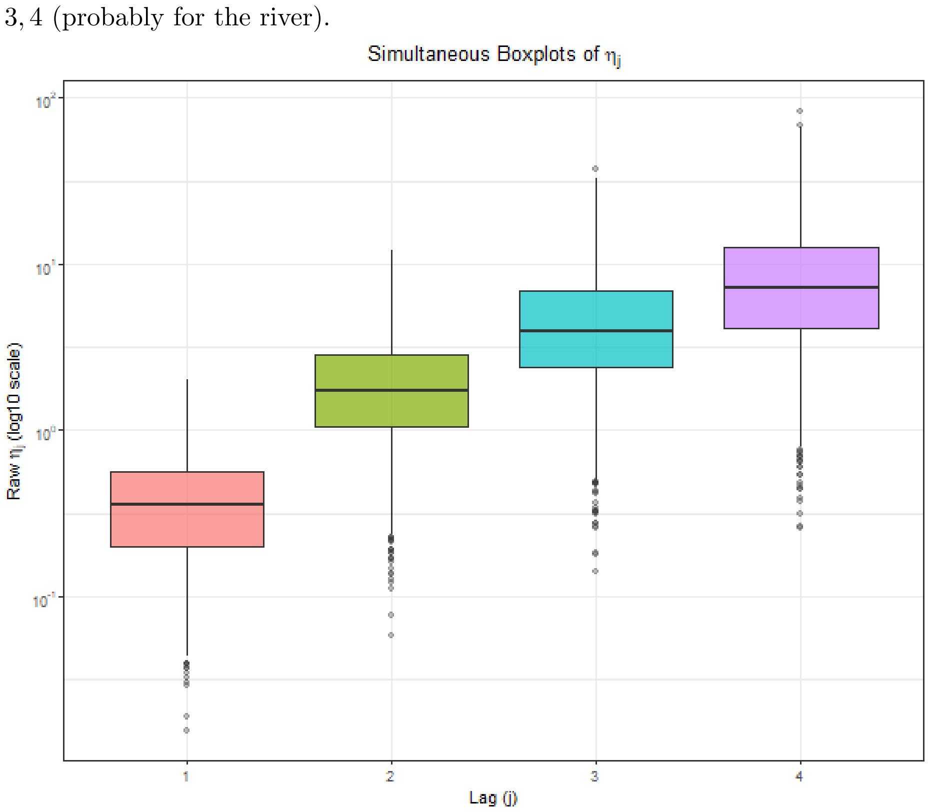
Left 2x2: local range for color gradient, right 2x2: global range for color gradient. In the left 2×2 panel I use a local color range for each lag, while in the right 2×2 panel I use a single global color range across all j. There is a similar pattern to the previous maps: with local scaling the $\log \eta_j$ values appear to have smaller magnitudes and the four panels look more alike, whereas with the global scale the full range of values is visible and differences between lags are easier to compare.

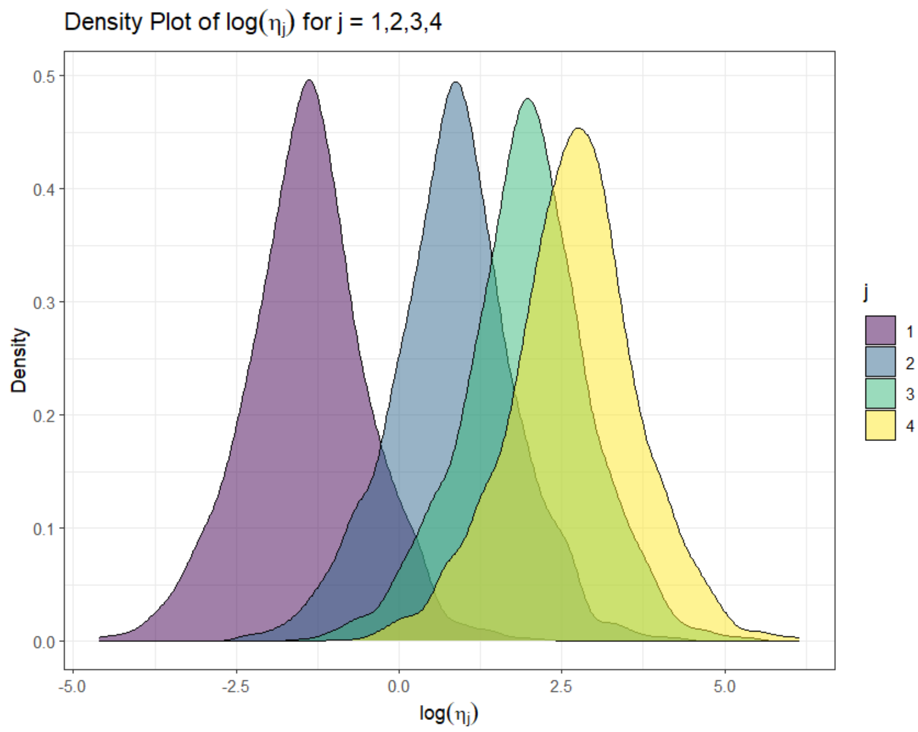
Density of $\log \eta_j$ For $j = 1, 2, 3, 4$



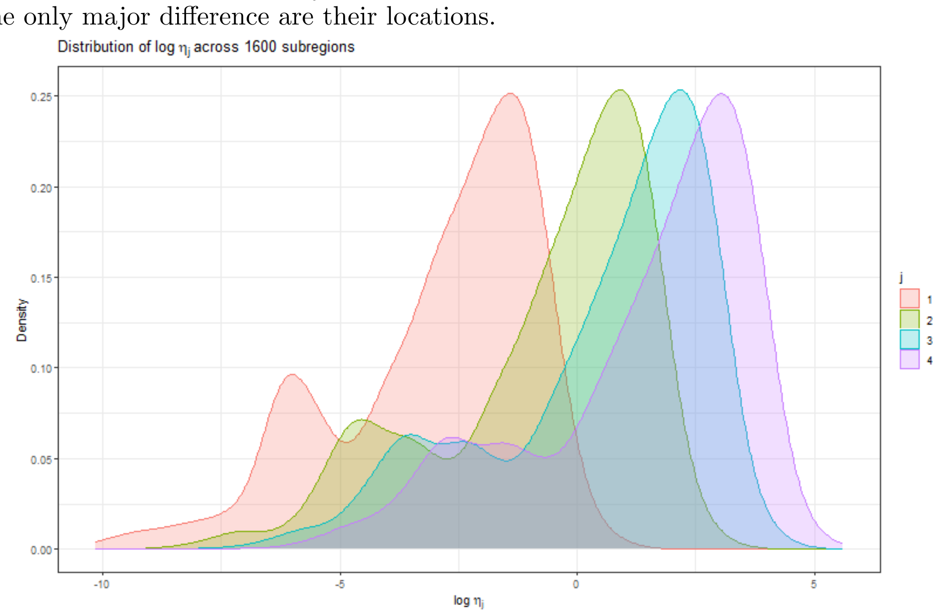
As j increases the distribution of $\log \eta_j$ shifts, which is expected because under the power law $\eta_j = cj^\kappa$, so $\log \eta_j = \log c + \kappa \log j$ which means $E[\log \eta_j] = \log c + \kappa \log j$. The actual amount the distributions shift by we can try to find by estimating κ with linear regression in each subregion (next plots).

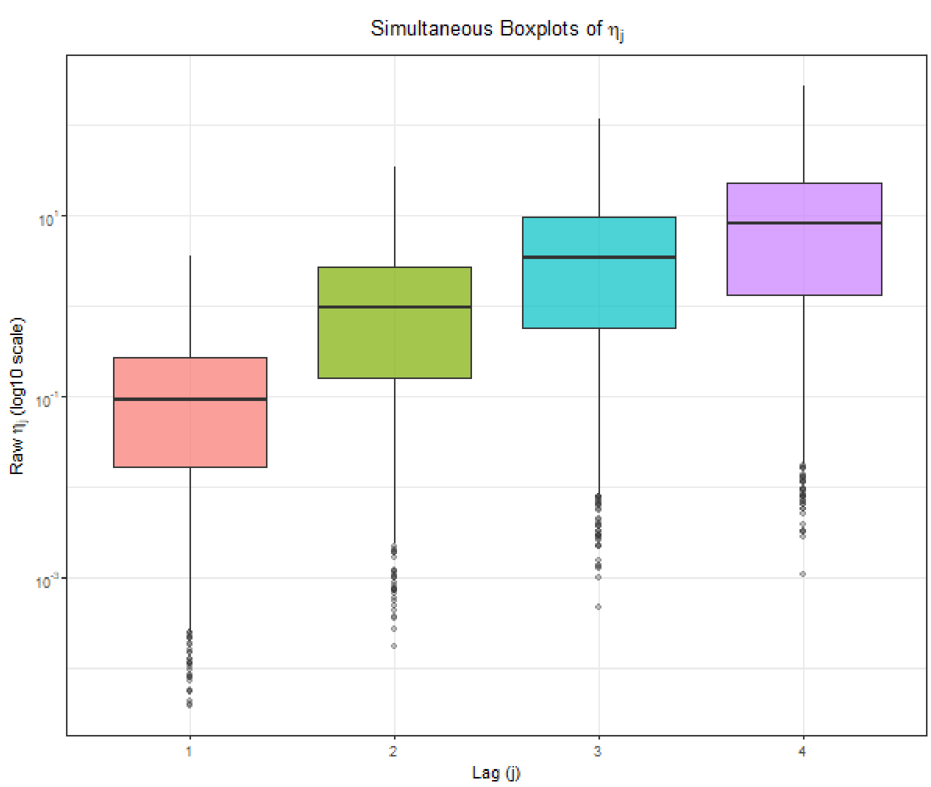
Something also note-worthy is that all the distributions are left-skewed, which clearly implies non-gaussianity. The following box plot shows this further with many outliers. There are also outliers above for $j = 3, 4$ (probably for the river).





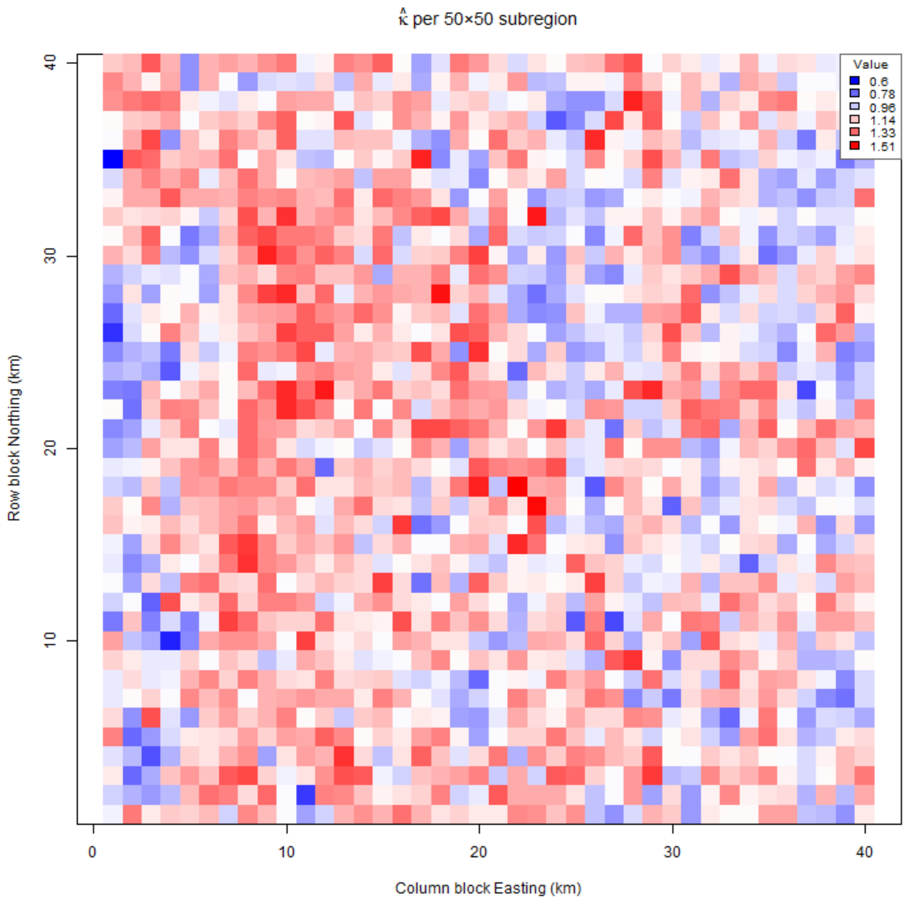
The distributions of $\log(\eta_j)$ for $j = 1, 2, 3, 4$ all look like they have similar shapes and are symmetric. The only major difference are their locations.



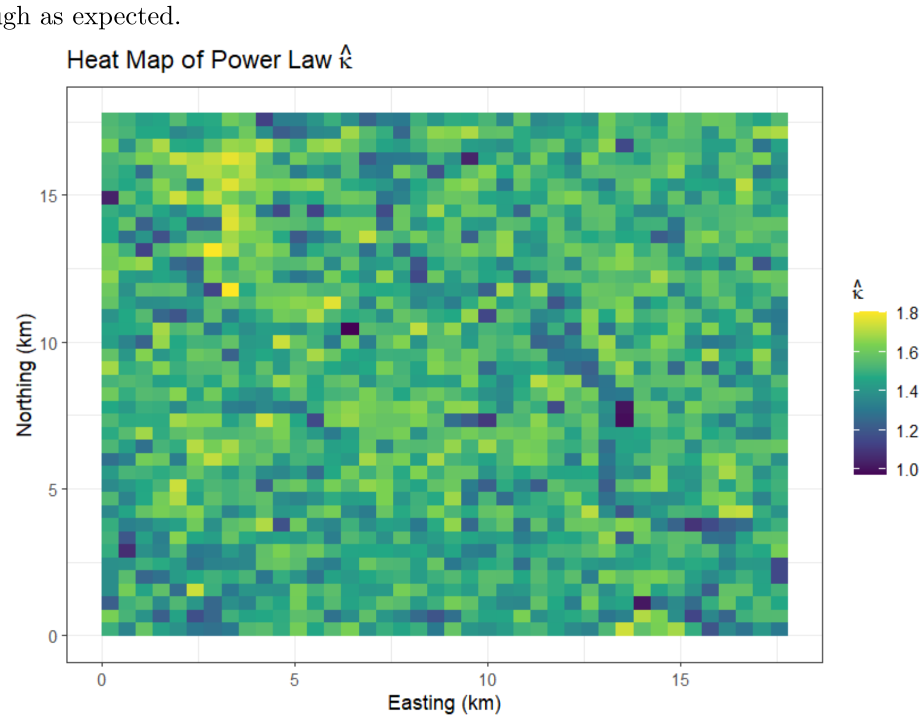


For my dataset, the density curves of $\log(\eta_j)$ for $j = 1, 2, 3, 4$ also have very similar overall shapes, and the main difference is a shift in location as j increases. The plots are very obviously non-Gaussian as they are very skewed, and there is a second smaller "hump" that is evening out as η_j gets larger

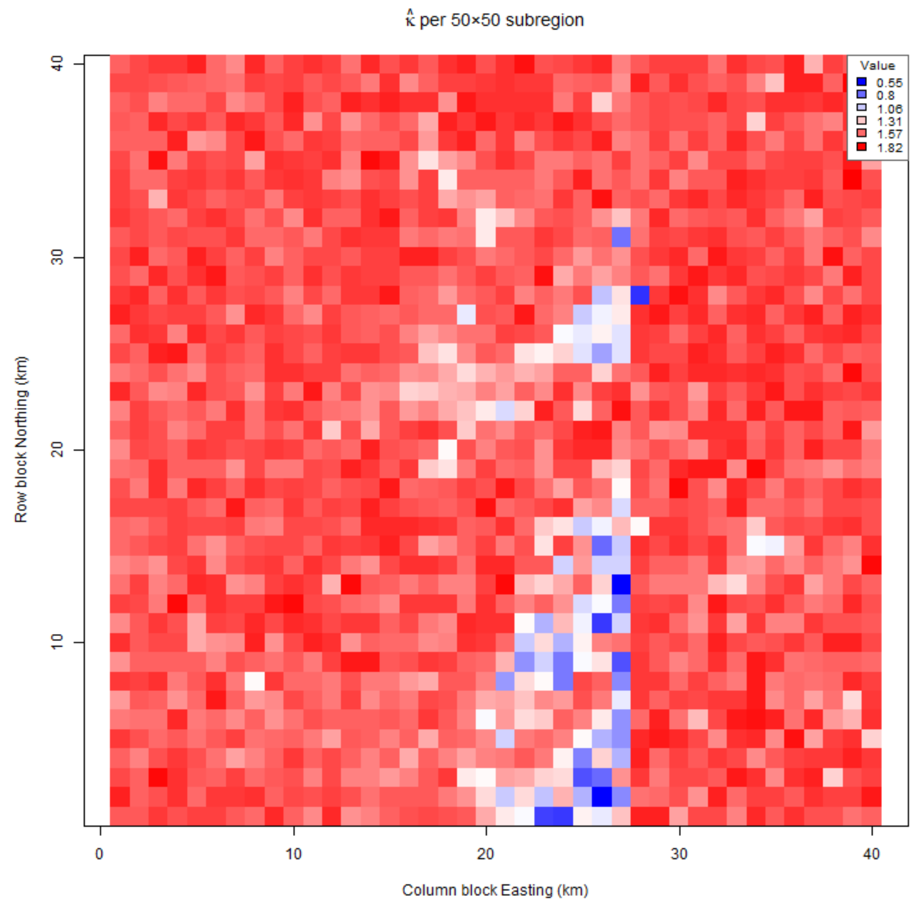
Heat Map Power-Law $\hat{\kappa}$ Estimates Per Subregion



The $\hat{\kappa}$ for each block is the smoothness for the power law model and were calculated using lm(log_eta ~ log_j) in r and taking the 2nd coefficient (κ) and dividing by 2 due to the prinipal irregular term. These $\hat{\kappa}$ do seem to match my previous analysis being between 0.6 to 1.51 smoothness overall.
This is a wide range and crosses differentiability $\kappa = 1$, which implies nonstationarity even further. There does seem to be a trend of higher powers near the left center of the map approaching the mountain, and partly on the peak of the mountain. There are also low $\hat{\kappa}$ around the far left and river where the blocks are rough as expected.

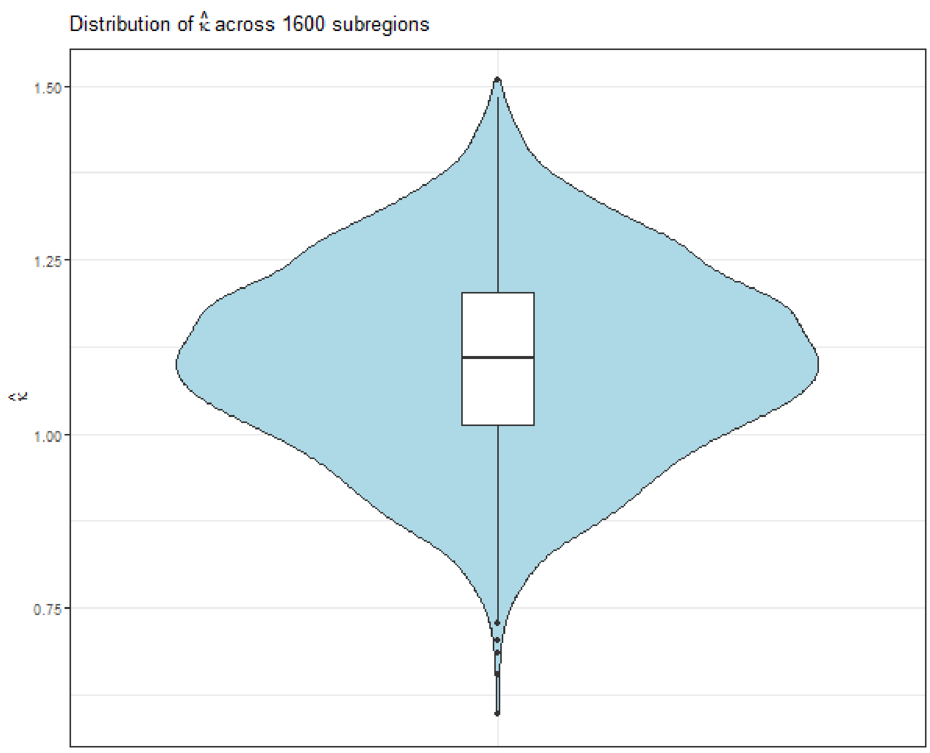


The smoothness for this region in the Poconos looks to be quite consistent sitting mostly between 1.4 and 1.8 in each sub region with a few regions being in the 1 to 1.4 range.

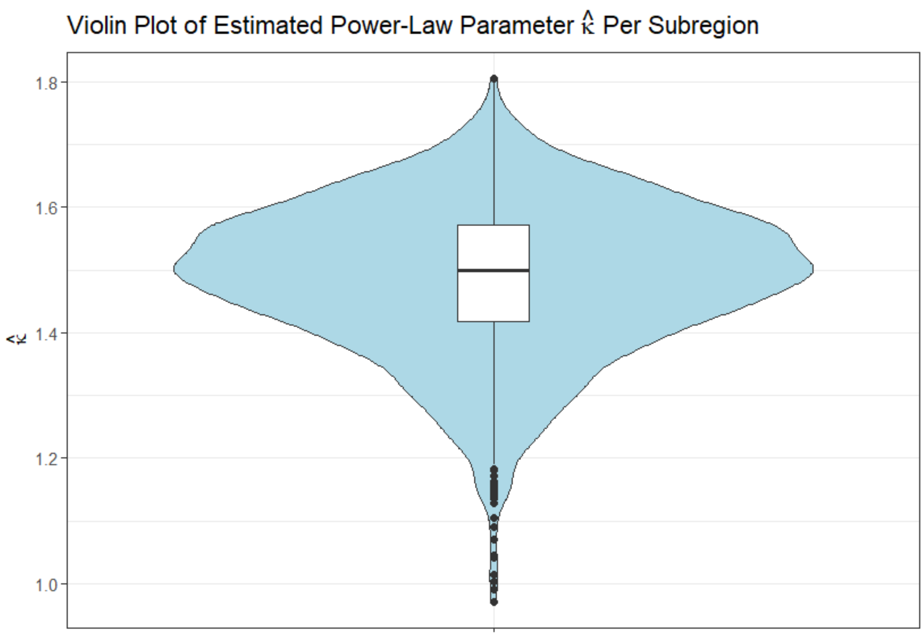


This plot shows how the majority of the subregions have a clear smoothness between 1 and 2, with some in the valleys having smoothness between .5 and 1. If there wasn't a dip in the smoothness in the middle/bottom area, there would be a better argument for stationarty. But looking at the contour map, I don't think that it is a reasonable assumption.

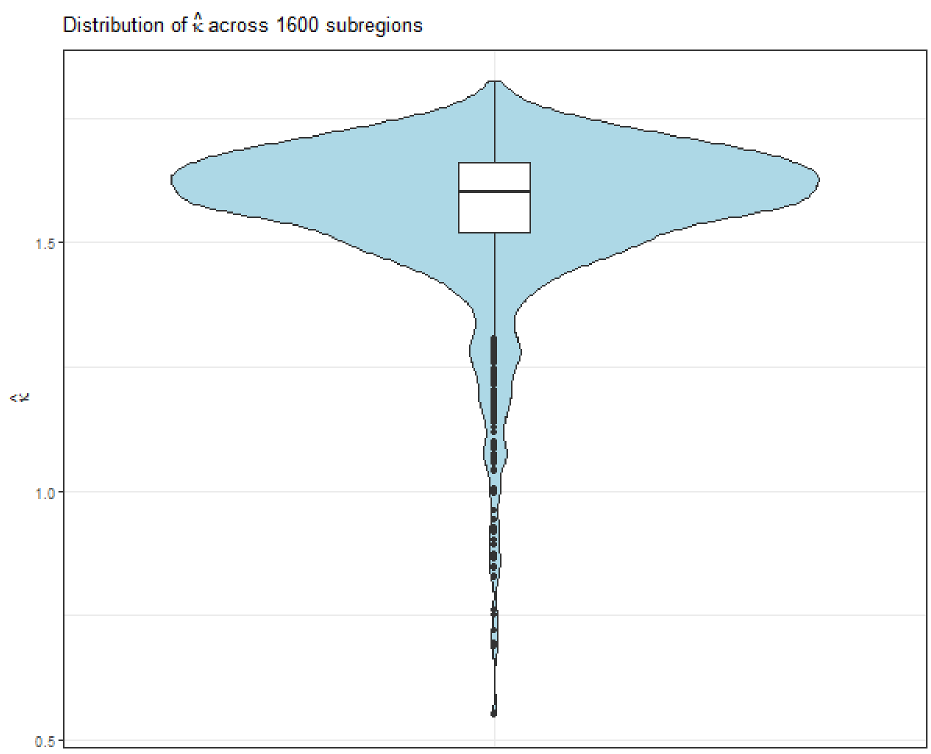
Distribution of Power-Law $\hat{\kappa}$ Estimates



The median in this violin plot is around 1.1, so the density plots shift by:
from $\log \eta_1$ to $\log \eta_2$: $1.1(\log 2 - \log(1)) = 0.76$
from $\log \eta_2$ to $\log \eta_3$: $1.1(\log 3 - \log 2)) = 0.45$
from $\log \eta_3$ to $\log \eta_4$: $1.1(\log 4 - \log 3) = 0.32$
and these shifts do seem consistent with the distribution plot.

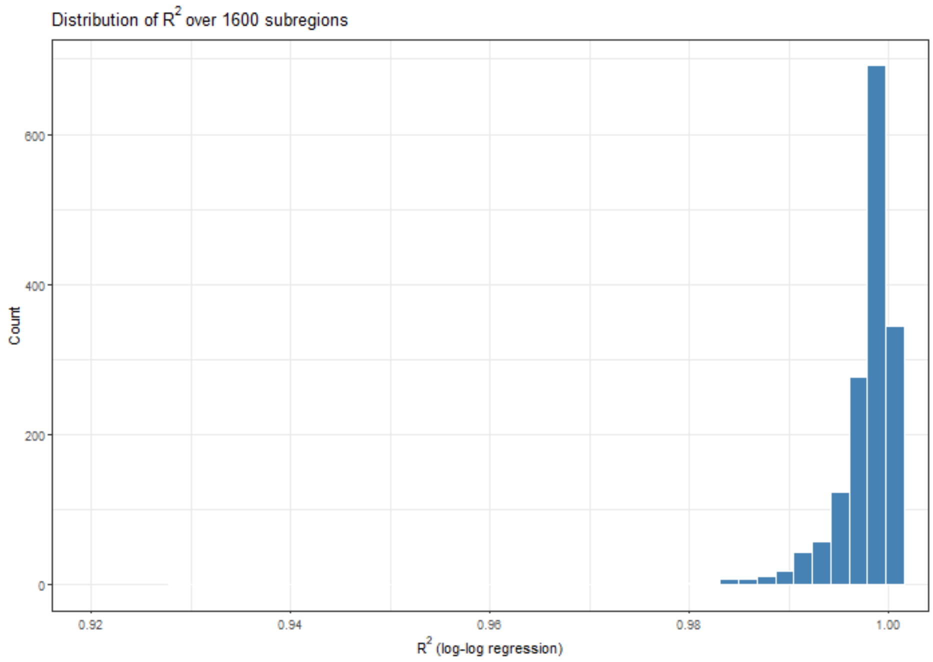


Similar to what was seen through the heat map, most of the estimates seem to be between 1.2 to 1.8 and the distribution is skewed to the left.

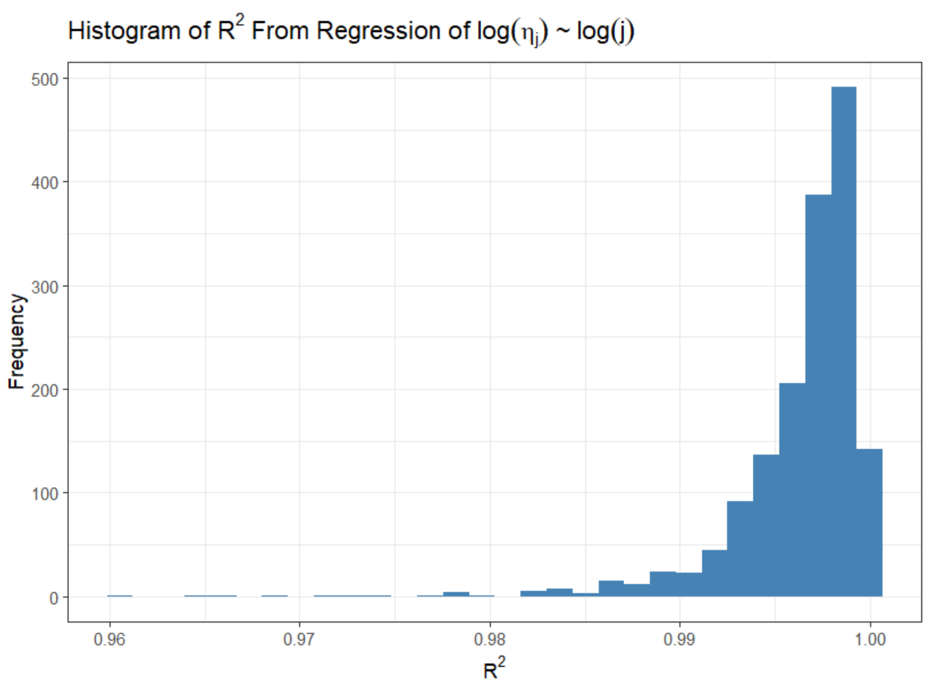


Extremely skewed plot. Smaller IQR than Luke's and George's plots, but there is a tail of kappa hats around .5 to 1.25.

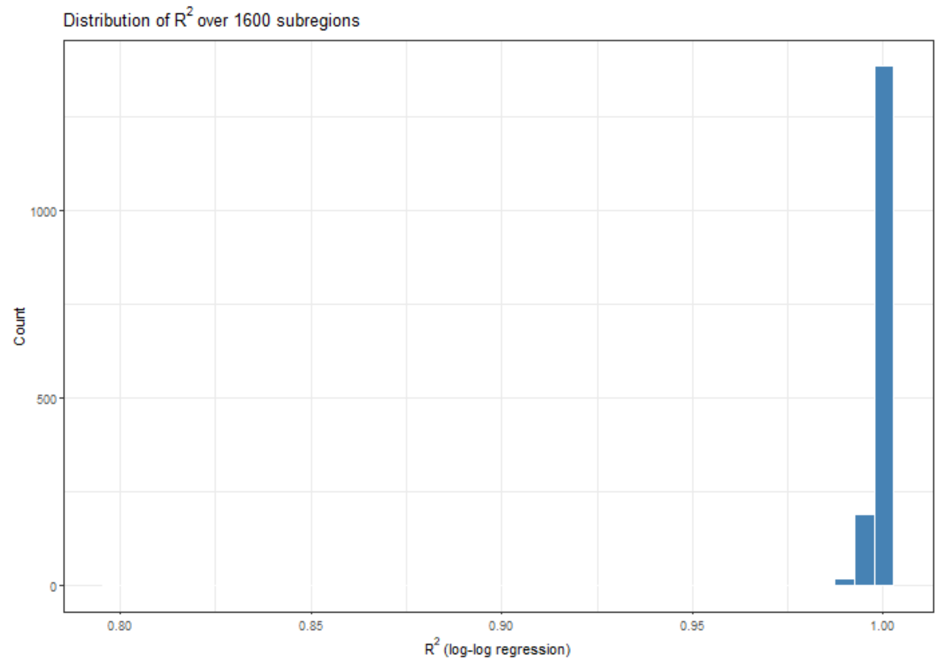
Distribution of R^2 From the Fitted Power-Law Model



There actually is some R^2 near the left of this plot (not all are clumped at 0.99). The lowest R^2 is at block row 37 column 3 at 0.927. This happens to also be a kappa 0.74, so it was a rough spot. But this does generally agree with a linear relationship between $\log \eta_j$ and $\log j$.

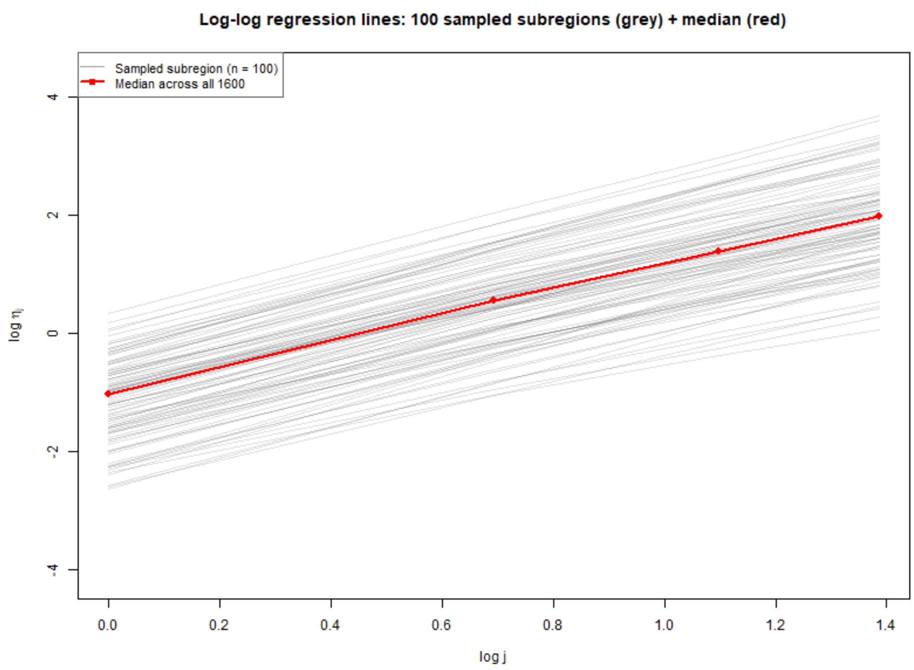


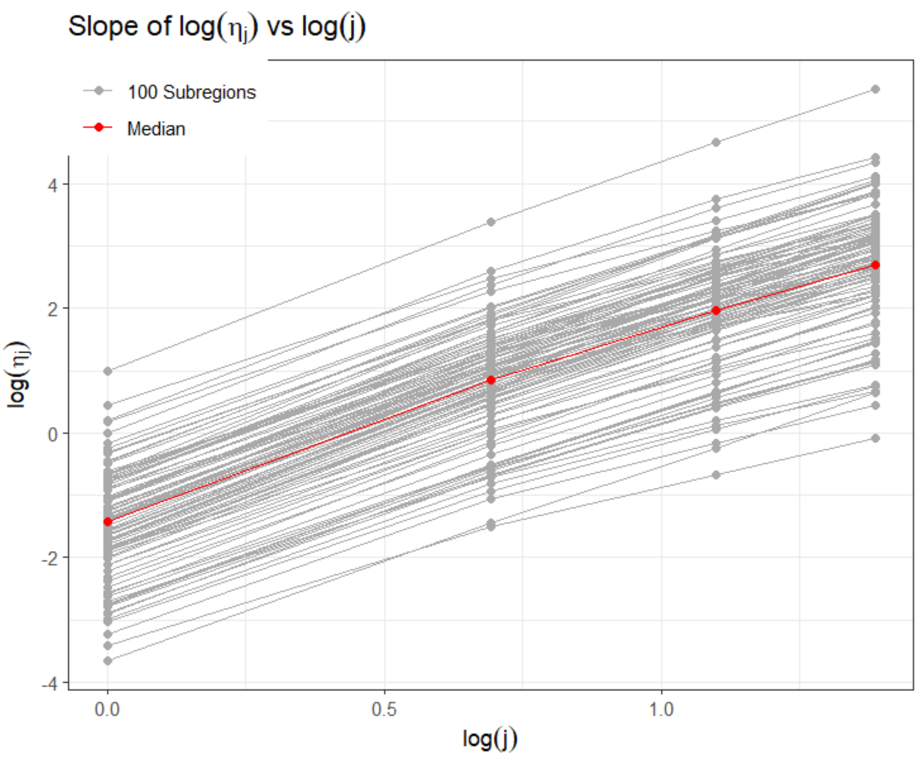
The R^2 values from the regression are all near 1, indicating that there is almost a perfect linear relationship between $\log(\eta_j)$ and $\log(j)$. The smallest R^2 seems to be about 0.96.



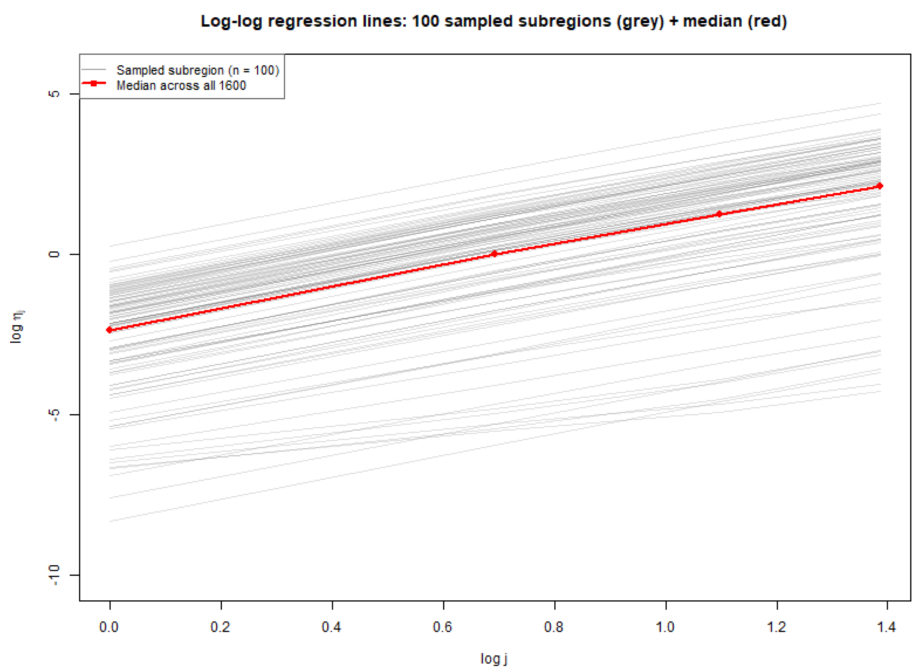
This plot does not help with much information, other than the fact that over 1000 of the R^2 values are close to 1. I fear this is concerning, but I am a little confused by it.

Relationship Between $\log \eta_j$ and $\log j$





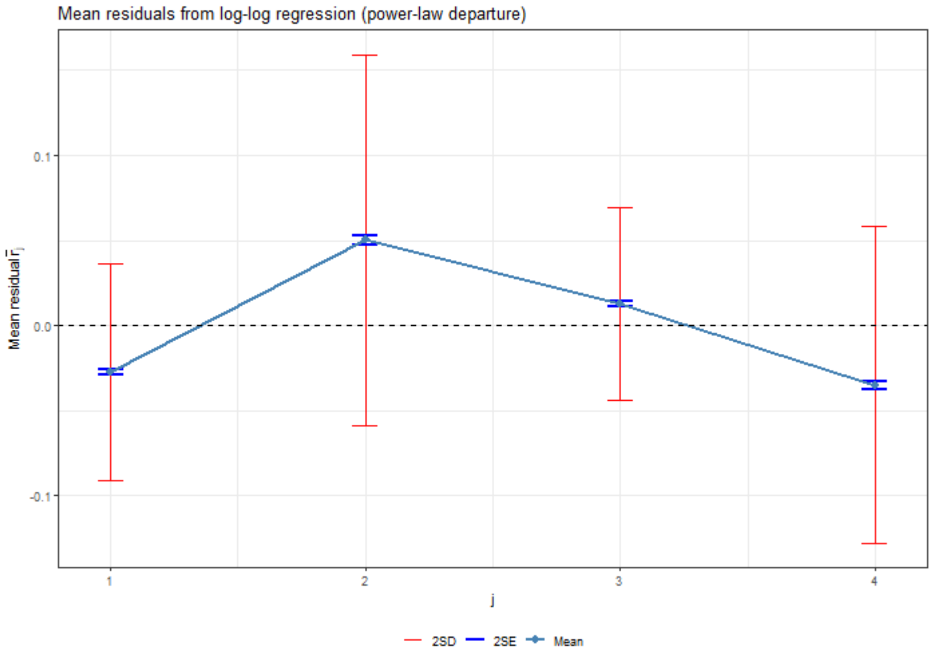
Even though the R^2 values are nearly 1, we can actually see some slight curvature in the relationship. However, it is very much so approximately linear which would indicate that the power-law is a good fit for this region.



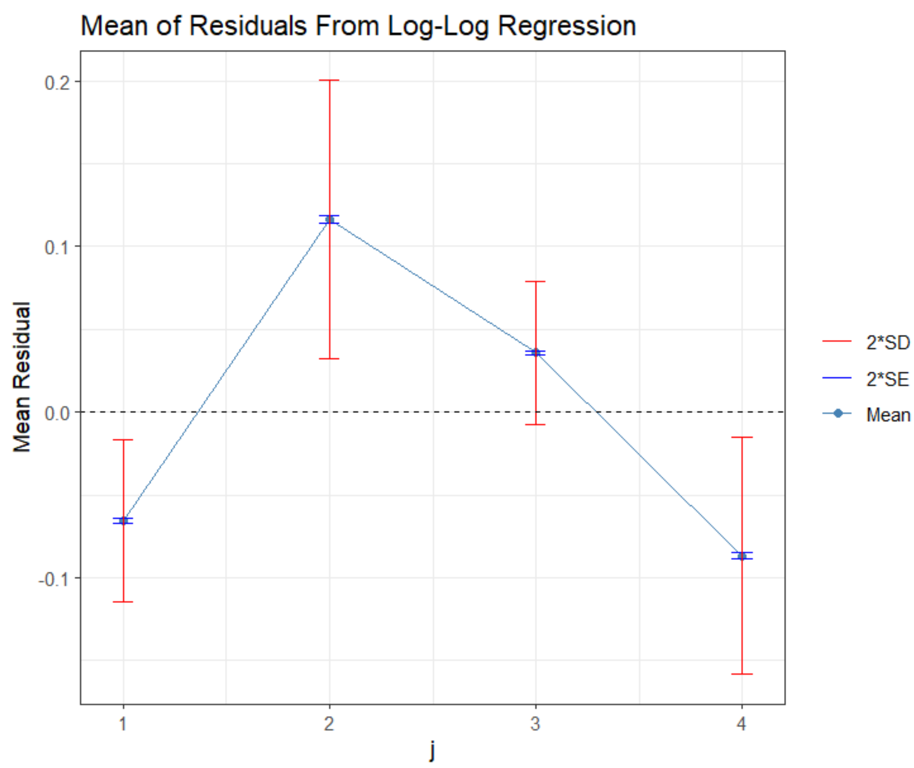
For my dataset, the overlaid lines of $\log(\eta_j)$ versus $\log(j)$ are also tightly clustered around a straight line, indicating a strong approximate power-law relationship within each block. It looks like the number of samples above the median are more tightly together, but that is expected given the prior plots

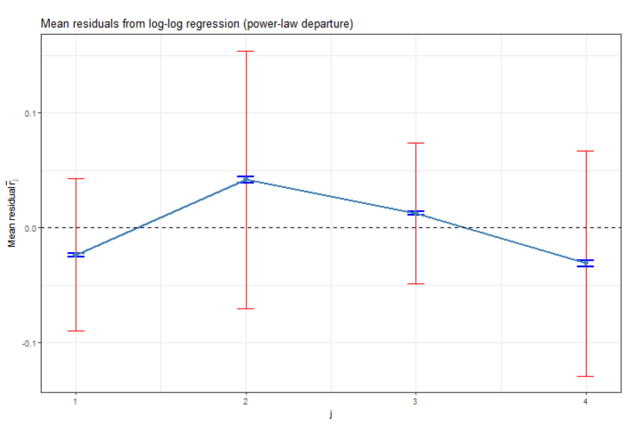
Mean of Residuals from $\log(\eta_j) \sim \log(j)$ Regression

For all three data sets, we see a systematic over prediction for $j = 1, 4$ and an under prediction for $j = 2, 3$.



resid_j1	resid_j2	resid_j3	resid_j4
-0.02763753	0.05003980	0.01261397	-0.03501623

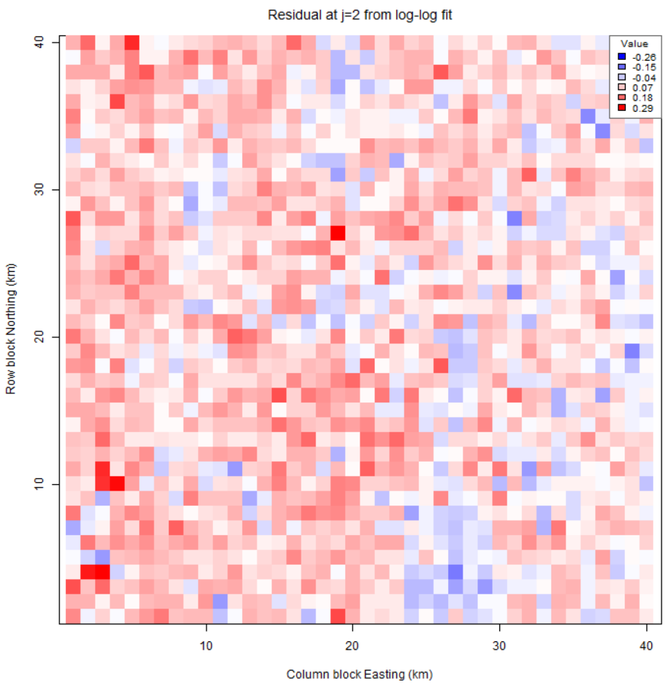


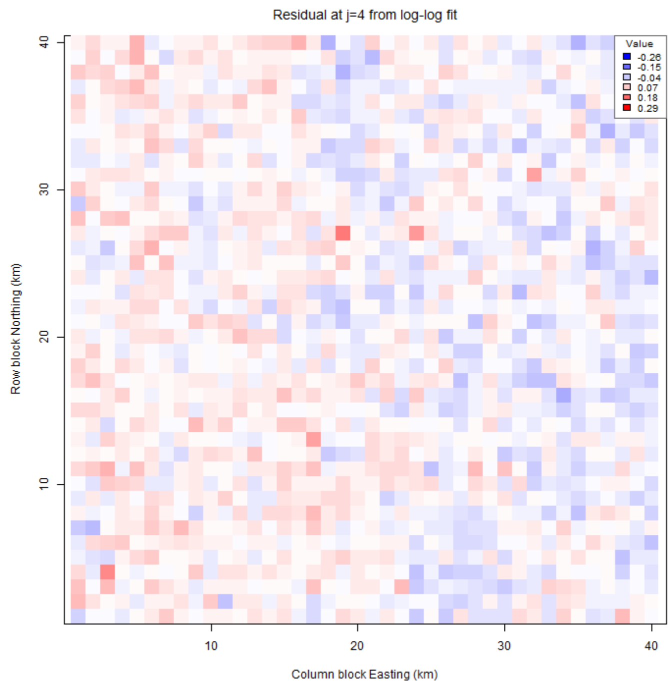


resid_j1	resid_j2	resid_j3	resid_j4
-0.02360013	0.04190830	0.01275055	-0.03105872

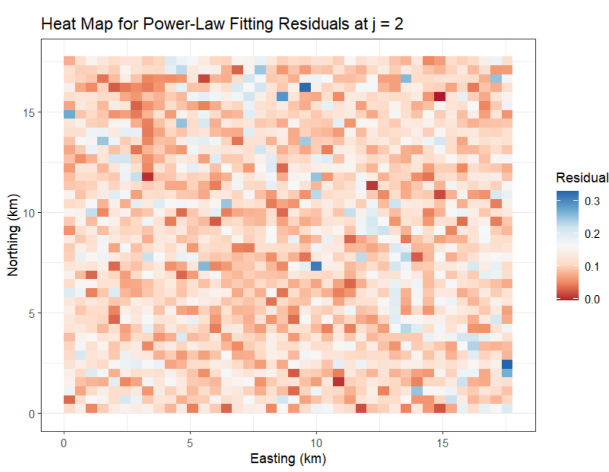
It looks like the residuals of the log-log regression are close to zero, where the $j = 1, 4$ overpredicts and $j = 2, 3$ underpredicts. There might be a small systematic curvature in the true relation of the model, but overall the fit is good.

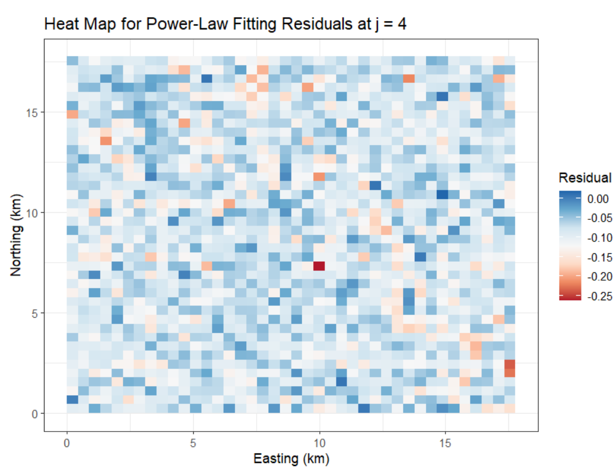
Heatmaps of Residuals at $j = 2$ and $j = 4$



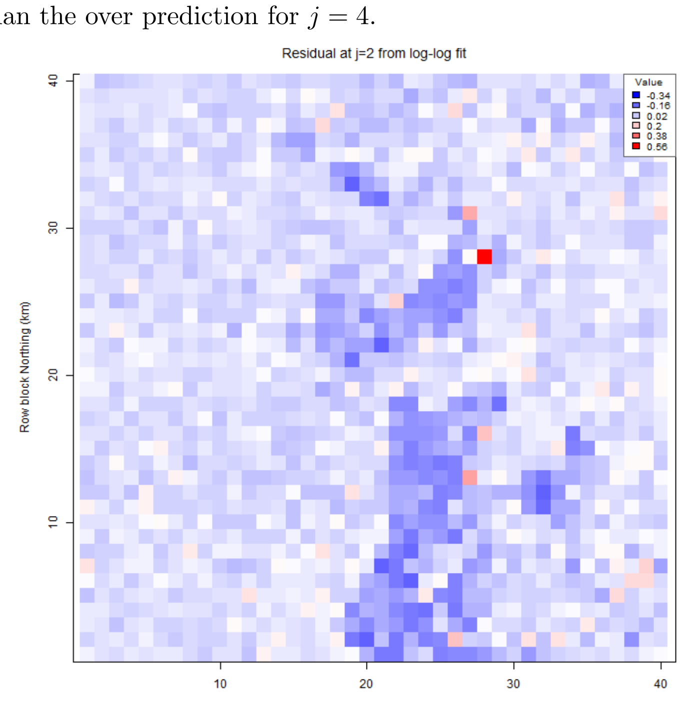


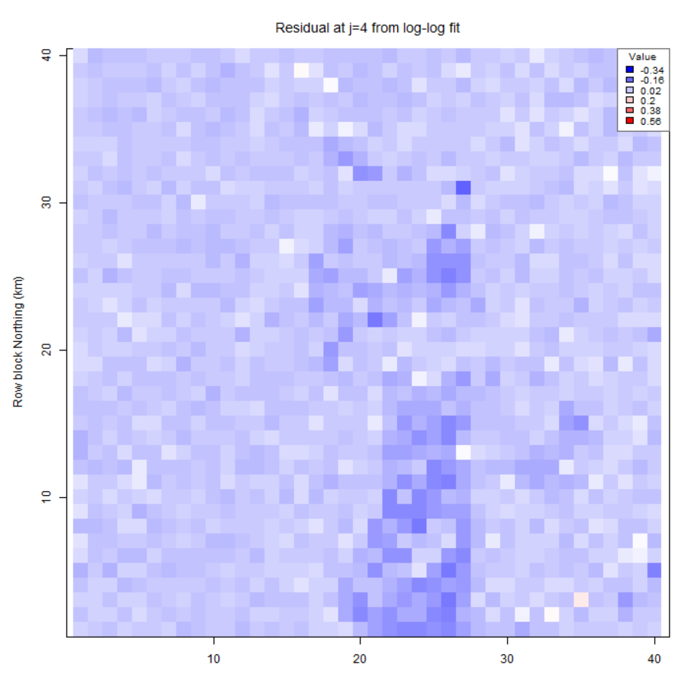
Here it looks like at $j = 2$ the residuals are mostly positive except along parts of the river. On the other hand $\log \eta_2$ residuals are mostly negative, showing that as lag increases we tend to under predict more with the power law in this terrain especially with the rough river.





For $j = 2$, there seems to be some clear under prediction going on near the river. For $j = 4$, we can sort of see an over prediction on the river but it's not as clear. The under prediction for $j = 2$ is clearly worse than the over prediction for $j = 4$.





It is clear that this model has major issues, as there is a clear pattern of the residuals in the valley areas. Also, most of the residuals are negative, so this shows that this model may not be a great fit for the data.